

Table 1: Selections and corrections applied during NanoAOD skimming, in execution order.

Step	Operation	Kind	Condition / definition	Scope	Correction state
1	Orthogonal luminosity split	Event filter	OrthogonalEraTag == 0/1/2 (seed 12345; luminosity fractions 110/111/26)	MC 2024–2026	Before all physics corrections
2	Generator VBF phase space	Definition only	Compute GenVBFFilter for selected inclusive/VBF-filtered DY samples; current code does not call df.Filter	MC DY 2024–2026	Before corrections; no rejection
3	Base event weights	Weight definition	Luminosity, generator sign, pileup and cross-section weights; configured QCD-scale sums	MC	Pileup is applied here; no rejection
4	Golden JSON	Event filter	Accept certified run:luminosityBlock pairs from maincfg.yaml	Data	Before object corrections
5	Muon ScaRe	Object correction	Momentum scale for data/MC; MC resolution smearing; NanoAOD and beam-spot-constrained momenta	Data and MC	Before every muon selection
6	Muon FSR recovery	Object correction	FSR photon recovery implemented in corrections/muon_fsr.py	Data and MC	After ScaRe; nominal selection uses Muon_pt_FSR_corr
7	JEC/JER	Object correction	Reapply JEC; apply JER to MC; produce JER/JES Total shifts when variations are enabled	Data and MC	Before MET and jet selections
8	MET quality flags	Event filter	Logical AND of goodVertices, globalSuperTightHalo2016Filter, EcalDeadCellTriggerPrimitiveFilter, BadPFMuonFilter, BadPFMuonDzFilter, hfNoisyHitsFilter, eeBadScFilter, ecalBadCalibFilter	Data and MC	NanoAOD flags; special 2022/2023 data bad-MET redefinition
9	Single-muon trigger matching	Event filter	HLT_IsoMu24; corrected+FSR muon pT > 26 GeV; trigger-object ID 13, filter bit 8, DeltaR < 0.4	Data and MC	After muon corrections; OR over nominal/shifts
10	Good muons	Object selection	corrected+FSR pT > 15.0 GeV; eta < 2.4; medium ID; pIsoId >= 2	Data and MC	ScaRe+FSR; nominal and muon shifts
11	Exactly two muons	Event filter	Exactly two good muons; nominal and every muon shift must pass when variations are enabled	Data and MC	After ScaRe, FSR and trigger matching
12	Dimuon mass window	Event filter	50 < m_mumu < 200 GeV; nominal and every muon shift must pass	Data and MC	Selected ScaRe+FSR muons
13	Electron veto	Event filter	No electron with pT > 20 GeV, eta < 2.5 and Electron_mvIso_WP90	Data and MC	Raw NanoAOD electron kinematics
14	Muon ID/ISO/trigger SF	Weight definition	Configured correctionlib SF sources for both selected muons; central and optional up/down	MC	After muon/electron filters; no rejection
15	Jet ID and b-tag flags	Object definition	Tight jet ID; b-tag flags at eta < 2.5; b-tag event-veto flag is stored but not filtered	Data and MC	JEC/JER jets and era-specific WPs
16	Jet veto map	Object definition	Evaluate map; apply_filter=False in analysis/skim.py	Data and MC	Corrected jets; no direct event filter
17	Selected jets	Object selection	corrected pT > 20.0 GeV; eta < 4.7; tight ID; outside veto map; DeltaR(j,mu1/2) > 0.4	Data and MC	JEC/JER jets; repeated for JER/JES shifts
18	VBF candidates	Definition only	Build pair outside the horn; store HasVBF and indices	Data and MC	No VBF event filter in skim
19	Snapshot	Output	Write surviving events and nominal/shifted columns	Data and MC	Final step; categories are downstream

Table 2: Era-dependent skim settings read from the YAML configuration.

Era	Muon/jet settings	Horn veto expression	Other event-filter settings
Run3_2022	mu pT>15.0 GeV, jet pT>20.0 GeV; eta_mu <2.4, eta_j <4.7	(abs(v_ops::eta(Jet_p4)) > 2.5 && v_ops::pt(Jet_p4) < 50.0)	trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24
Run3_2022EE	mu pT>15.0 GeV, jet pT>20.0 GeV; eta_mu <2.4, eta_j <4.7	(abs(v_ops::eta(Jet_p4)) >= 2.5 && v_ops::pt(Jet_p4) < 50.0)	trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24; special bad-MET runs=362433–362760
Run3_2023	mu pT>15.0 GeV, jet pT>20.0 GeV; eta_mu <2.4, eta_j <4.7	(abs(v_ops::eta(Jet_p4)) >= 2.5 && v_ops::pt(Jet_p4) < 50.0)	trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24; special bad-MET runs=366727–367144
Run3_2023BPix	mu pT>15.0 GeV, jet pT>20.0 GeV; eta_mu <2.4, eta_j <4.7	(abs(v_ops::eta(Jet_p4)) >= 2.5 && v_ops::pt(Jet_p4) < 50.0)	trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24
Run3_2024	mu pT>15.0 GeV, jet pT>20.0 GeV; eta_mu <2.4, eta_j <4.7	((abs(v_ops::eta(Jet_p4)) >= 2.5 && abs(v_ops::eta(Jet_p4)) < 3.0) && v_ops::pt(Jet_p4) < 50.0)	trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24
Run3_2025	mu pT>15.0 GeV, jet pT>20.0 GeV; eta_mu <2.4, eta_j <4.7	(abs(v_ops::eta(Jet_p4)) < 0)	trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24
Run3_2026	mu pT>15.0 GeV, jet pT>20.0 GeV; eta_mu <2.4, eta_j <4.7	abs(v_ops::eta(Jet_p4)) < 0	trigger filter=yes; jet-veto event filter=no; HLT=HLT_IsoMu24

Important scope note. The muons_selection, categories, and masses_regions expressions in config/Run3_*/selections.yaml are not invoked by the current analysis/skim.py. They are downstream histogram-level definitions, not NanoAOD skim event filters.

Table 3: Histogram-production workflow in execution order.

Step	Operation	Action	Selection/correction status
1	Input skim chunks	Read validated Events trees; MC normalization denominators come from dataset-wide skim reports/manifest metadata	No new object calibration; consumes corrected and shifted skim branches
2	Optional additional cut	If <code>–additional-cuts</code> is supplied, apply it directly with <code>RDataFrame.Filter</code>	User-defined event filter, before derived observables and weights
3	Weights and observables	Build central/shifted weights, trigger SF, PDF and QCD-scale weights; define selected-jet, VBF, dimuon, soft-activity and requested DNN observables	Uses corrections stored by the skim
4	Shifted jet observables	For requested JER/JES suffixes, rebuild selected-jet and VBF observables; fail if required skim columns are absent	Shape-systematic preparation
5	Reco/gen jet matching	For eligible MC, define jet matching and optional DY/EWK exclusive jet-component categories	MC only; optional component workflow
6	Histogram selections	Define/redefine every <code>masses_regions</code> , <code>muons_selection</code> , <code>jets_selection</code> and categories expression from <code>selections.yaml</code> , including requested shape suffixes	Definitions first; no rejection yet
7	DY normalization/reweights	Apply <code>amcatnlo</code> normalization and optional DY pT(l) and Njets reweights to all target weight columns	MC DY only; weights, not filters
8	Mass-region filter	Filter on each requested stored mass-region column (default: <code>mass_inclusive</code> , <code>Z_sideband</code> , <code>Signal_Fit</code>)	First histogram-level event selection
9	Sideband DNN	For DNN_NNOutput in Z/H sidebands, remap <code>m_mumu</code> into the training window and reevaluate the dedicated DNN payload	Only requested DNN sideband histograms
10	Category filter	Within each mass region, filter each requested stored category (default: <code>baseline</code> , <code>ggF</code> , <code>VBF</code>)	Second histogram-level event selection
11	Systematic routing	Choose shape-shifted variable/selection columns or weight-only variations; data are forced to Central	JER/JES/Muon scale/resolution shapes; PU/muon/PDF/QCD weights
12	Histogram booking	Book configured 1D/2D models with the selected variable and weight; missing shifted variables produce empty templates	Output organized by mass region and category
13	Chunk merge/postprocessing	Merge temporary ROOT outputs; build QCD scale envelopes, PDF treatment and optional split component files	Final histogram products

Table 4: All histogram-level selection definitions. Rows are merged when identical across eras.

YAML section	Name	Expression	Store	Eras	Histogram use
<code>masses_regions</code>	<code>Z_sideband</code>	<code>m_mumu{mu_suff} > 70 && m_mumu{mu_suff} < 110</code>	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
<code>masses_regions</code>	<code>Signal_Fit</code>	<code>m_mumu{mu_suff} > 115 && m_mumu{mu_suff} < 135</code>	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
<code>masses_regions</code>	<code>H_sideband</code>	<code>((m_mumu{mu_suff} > 110 && m_mumu{mu_suff} < 115) (m_mumu{mu_suff} > 135 && m_mumu{mu_suff} < 150))</code>	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
<code>masses_regions</code>	<code>Signal_ext</code>	<code>m_mumu{mu_suff} > 110 && m_mumu{mu_suff} < 150</code>	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
<code>masses_regions</code>	<code>mass_inclusive</code>	<code>m_mumu{mu_suff} > 60 && m_mumu{mu_suff} < 150</code>	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
<code>muons_selection</code>	<code>muon_pre_sel_check</code>	<code>mu1_pt{mu_suff} > 15 && abs(mu1_eta) < 2.4 && mu2_pt{mu_suff} > 15 && abs(mu2_eta) < 2.4</code>	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
<code>muons_selection</code>	<code>muons_OS</code>	<code>mu1_charge{mu_suff} * mu2_charge{mu_suff} < 0</code>	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
<code>muons_selection</code>	<code>muons_OS_and_presel</code>	<code>muon_pre_sel_check{mu_suff} && muons_OS{mu_suff}</code>	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
<code>muons_selection</code>	<code>muons_trigger</code>	<code>(Event_HasTriggerMatching_singleMu{mu_suff} && HLT_IsoMu24 && ((mu1_pt{mu_suff} > 26 && mu1_HasTriggerMatching_singleMu{mu_suff}) (mu2_pt{mu_suff} > 26 && mu2_HasTriggerMatching_singleMu{mu_suff})))</code>	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
<code>muons_selection</code>	<code>muons_OS_presel_trg</code>	<code>muons_OS_and_presel{mu_suff} && muons_trigger{mu_suff}</code>	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
<code>muons_selection</code>	<code>mu1_pt_l30</code>	<code>mu1_pt{mu_suff} < 30</code>	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
<code>muons_selection</code>	<code>mu2_pt_l30</code>	<code>mu2_pt{mu_suff} < 30</code>	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
<code>muons_selection</code>	<code>mu1_pt_g20</code>	<code>mu1_pt{mu_suff} > 20</code>	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
<code>muons_selection</code>	<code>mu2_pt_g20</code>	<code>mu2_pt{mu_suff} > 20</code>	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
<code>muons_selection</code>	<code>mu_pt_sel</code>	<code>mu1_pt_g20{mu_suff} && mu2_pt_g20{mu_suff}</code>	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
<code>muons_selection</code>	<code>mu1_ML</code>	<code>mu1_pflIsoId{mu_suff} >= 2 && mu1_mediumId{mu_suff}</code>	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
<code>muons_selection</code>	<code>mu1_LL</code>	<code>mu1_pflIsoId{mu_suff} >= 2 && mu1_looseId{mu_suff}</code>	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
<code>muons_selection</code>	<code>mu1_TT</code>	<code>mu1_pflIsoId{mu_suff} >= 4 && mu1_tightId{mu_suff}</code>	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
<code>muons_selection</code>	<code>mu2_ML</code>	<code>mu2_pflIsoId{mu_suff} >= 2 && mu2_mediumId{mu_suff}</code>	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
<code>muons_selection</code>	<code>mu2_LL</code>	<code>mu2_pflIsoId{mu_suff} >= 2 && mu2_looseId{mu_suff}</code>	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
<code>muons_selection</code>	<code>mu2_TT</code>	<code>mu2_pflIsoId{mu_suff} >= 4 && mu2_tightId{mu_suff}</code>	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
<code>muons_selection</code>	<code>bothMuons_ML</code>	<code>mu1_ML{mu_suff} && mu2_ML{mu_suff}</code>	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency

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Table 4 – continued

YAML section	Name	Expression	Store	Eras	Histogram use
muons_selection	bothMuons_LL	mu1_LL{mu_suff} && mu2_LL{mu_suff}	no	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
muons_selection	chi2_sel	mu1_bsConstrainedChi2{mu_suff} < 30 && mu2_bsConstrainedChi2{mu_suff} < 30	yes	2022, 2022EE, 2023, 2023BPix, 2025, 2026	definition/dependency
muons_selection	pt_less than30_and_TT	((mu1_pt_l30{mu_suff} && mu1_TT{mu_suff}) !(mu1_pt_l30{mu_suff})) && ((mu2_pt_l30{mu_suff} && mu2_TT{mu_suff}) !(mu2_pt_l30{mu_suff})))	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
jets_selection	has_good_jets	N_SelectedJets{jet_suff} >= 0	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
jets_selection	jets_btag_clean_sel	SelectedJetTagSel{jet_suff}	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	definition/dependency
categories	base_sel	muons_OS_presel_trg{mu_suff} && mu_pt_sel{mu_suff} && SelectedJetTagSel{jet_suff}	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
categories	baseline	base_sel{tot_suff} && mu1_ML{mu_suff} && mu2_ML{mu_suff}	yes	2022, 2022EE, 2023, 2023BPix, 2026	filter candidate
categories	baseline_chi2	baseline{tot_suff} && chi2_sel{mu_suff}	yes	2022, 2022EE, 2023, 2023BPix, 2025, 2026	filter candidate
categories	baseline_lowPtTT	baseline{tot_suff} && pt_less than30_and_TT{mu_suff}	yes	2022, 2022EE, 2023, 2023BPix, 2026	filter candidate
categories	baseline_chi2_lowPtTT	baseline{tot_suff} && pt_less than30_and_TT{mu_suff} && chi2_sel{mu_suff}	yes	2022, 2022EE, 2023, 2023BPix, 2026	filter candidate
categories	VBF_def	HasVBF{jet_suff} && SelectedJet_pt{jet_suff}[VBFJetIdx_1{jet_suff}] >= 35 && SelectedJet_pt{jet_suff}[VBFJetIdx_2{jet_suff}] >= 25	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
categories	VBF	baseline{tot_suff} && VBF_def{tot_suff}	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
categories	ggF	baseline{tot_suff} && !(VBF_def{tot_suff})	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
categories	ggF_0J	ggF{tot_suff} && N_SelectedJets{jet_suff} == 0	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
categories	ggF_1J	ggF{tot_suff} && N_SelectedJets{jet_suff} == 1	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
categories	ggF_ge2J	ggF{tot_suff} && N_SelectedJets{jet_suff} >= 2	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
categories	VBF_ge2J	VBF{tot_suff} && N_SelectedJets{jet_suff} >= 2	yes	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026	filter candidate
categories	VBF_lowPtTT	baseline_lowPtTT{tot_suff} && VBF_def{tot_suff}	yes	2022	filter candidate
categories	ggF_lowPtTT	baseline_lowPtTT{tot_suff} && !(VBF_def{tot_suff})	yes	2022	filter candidate
categories	ggF_Z_0J	ggF_0J{tot_suff}	yes	2022EE, 2023, 2023BPix, 2025, 2026	filter candidate
categories	ggF_Z_1J	ggF_1J{tot_suff}	yes	2022EE, 2023, 2023BPix, 2025, 2026	filter candidate
categories	ggF_Z_ge2J	ggF_ge2J{tot_suff}	yes	2022EE, 2023, 2023BPix, 2025, 2026	filter candidate
categories	VBF_Z_ge2J	VBF_ge2J{tot_suff}	yes	2022EE, 2023, 2023BPix, 2025, 2026	filter candidate
categories	baseline	base_sel{tot_suff} && mu1_ML{mu_suff} && mu2_ML{mu_suff} && pt_less than30_and_TT{mu_suff}	yes	2024, 2025	filter candidate

Table 5: Shape and weight systematics configured for histogram production.

Type	Systematic	Muon suffix	Jet suffix / weight expression	Weight column	Eras
shape	Central			weight__Central	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026
shape	JER		_JER{scale}	weight__Central	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026
shape	JES_Total		_JESTotal{scale}	weight__Central	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026
shape	MuonScale	_FSR_scale_{scale}		weight__Central	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026
shape	MuonRes	_FSR_res_{scale}		weight__Central	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026
weight	Central		genWeight * weight_lumi * weight_xs * weight_pu * weight_mu1_LoosePFIso_MediumID_Central * weight_mu1_MediumID_Trk_Central * weight_mu2_LoosePFIso_MediumID_Central * weight_mu2_MediumID_Trk_Central * weight_TrgSF_singleMu_IsoMu24Central	weight__Central	2022, 2022EE, 2023, 2023BPix

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Table 5 – continued

Type	Systematic	Muon suffix	Jet suffix / weight expression	Weight column	Eras
weight	QCDScale_{scale}		derived/envelope	weight__QCDScale_{scale}	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026
weight	PDF_{scale}		weight_pdf_{scale}	weight__PDF_{scale}	2022, 2022EE, 2023, 2023BPix, 2024, 2025, 2026
weight	PU_{scale}		genWeight * weight_lumi * weight_xs * weight_pu_{scale} * weight_mu1_LoosePFIso_MediumID_Central * weight_mu1_MediumID_Trk_Central * weight_mu2_LoosePFIso_MediumID_Central * weight_mu2_MediumID_Trk_Central * weight_TrqSF_singleMu_IsoMu24Central	weight__PU_{scale}	2022, 2022EE, 2023, 2023BPix
weight	MuonID_{scale}		genWeight * weight_lumi * weight_xs * weight_pu * weight_mu1_LoosePFIso_MediumID_Central * weight_mu1_MediumID_Trk_{scale} * weight_mu2_LoosePFIso_MediumID_Central * weight_mu2_MediumID_Trk_{scale} * weight_TrqSF_singleMu_IsoMu24Central	weight__MuonID_{scale}	2022, 2022EE, 2023, 2023BPix
weight	MuonIso_{scale}		genWeight * weight_lumi * weight_xs * weight_pu * weight_mu1_LoosePFIso_MediumID_{scale} * weight_mu1_MediumID_Trk_Central * weight_mu2_LoosePFIso_MediumID_{scale} * weight_mu2_MediumID_Trk_Central * weight_TrqSF_singleMu_IsoMu24Central	weight__MuonIso_{scale}	2022, 2022EE, 2023, 2023BPix
weight	singleMuTrigger_{scale}		genWeight * weight_lumi * weight_xs * weight_pu * weight_mu1_LoosePFIso_MediumID_Central * weight_mu1_MediumID_Trk_Central * weight_mu2_LoosePFIso_MediumID_Central * weight_mu2_MediumID_Trk_Central * weight_TrqSF_singleMu_IsoMu24{scale}	weight__singleMuTrigger_{scale}	2022, 2022EE, 2023, 2023BPix
weight	Central		genWeight * weight_lumi * weight_xs * weight_pu * ((mu1_pt < 30) ? weight_mu1_TightPFIso_TightID_Central : weight_mu1_LoosePFIso_MediumID_Central) * ((mu1_pt < 30) ? weight_mu1_TightID_Trk_Central : weight_mu1_MediumID_Trk_Central) * ((mu2_pt < 30) ? weight_mu2_TightPFIso_TightID_Central : weight_mu2_LoosePFIso_MediumID_Central) * ((mu2_pt < 30) ? weight_mu2_TightID_Trk_Central : weight_mu2_MediumID_Trk_Central) * weight_TrqSF_singleMu_IsoMu24Central	weight__Central	2024, 2025
weight	PU_{scale}		genWeight * weight_lumi * weight_xs * weight_pu_{scale} * ((mu1_pt < 30) ? weight_mu1_TightPFIso_TightID_Central : weight_mu1_LoosePFIso_MediumID_Central) * ((mu1_pt < 30) ? weight_mu1_TightID_Trk_Central : weight_mu1_MediumID_Trk_Central) * ((mu2_pt < 30) ? weight_mu2_TightPFIso_TightID_Central : weight_mu2_LoosePFIso_MediumID_Central) * ((mu2_pt < 30) ? weight_mu2_TightID_Trk_Central : weight_mu2_MediumID_Trk_Central) * weight_TrqSF_singleMu_IsoMu24Central	weight__PU_{scale}	2024, 2025
weight	MuonID_{scale}		genWeight * weight_lumi * weight_xs * weight_pu * ((mu1_pt < 30) ? weight_mu1_TightPFIso_TightID_Central : weight_mu1_LoosePFIso_MediumID_Central) * ((mu1_pt < 30) ? weight_mu1_TightID_Trk_{scale} : weight_mu1_MediumID_Trk_{scale}) * ((mu2_pt < 30) ? weight_mu2_TightPFIso_TightID_Central : weight_mu2_LoosePFIso_MediumID_Central) * ((mu2_pt < 30) ? weight_mu2_TightID_Trk_{scale} : weight_mu2_MediumID_Trk_{scale}) * weight_TrqSF_singleMu_IsoMu24Central	weight__MuonID_{scale}	2024, 2025
weight	MuonIso_{scale}		genWeight * weight_lumi * weight_xs * weight_pu * ((mu1_pt < 30) ? weight_mu1_TightPFIso_TightID_{scale} : weight_mu1_LoosePFIso_MediumID_{scale}) * ((mu1_pt < 30) ? weight_mu1_TightID_Trk_Central : weight_mu1_MediumID_Trk_Central) * ((mu2_pt < 30) ? weight_mu2_TightPFIso_TightID_{scale} : weight_mu2_LoosePFIso_MediumID_{scale}) * ((mu2_pt < 30) ? weight_mu2_TightID_Trk_Central : weight_mu2_MediumID_Trk_Central) * weight_TrqSF_singleMu_IsoMu24Central	weight__MuonIso_{scale}	2024, 2025
weight	singleMuTrigger_{scale}		genWeight * weight_lumi * weight_xs * weight_pu * ((mu1_pt < 30) ? weight_mu1_TightPFIso_TightID_Central : weight_mu1_LoosePFIso_MediumID_Central) * ((mu1_pt < 30) ? weight_mu1_TightID_Trk_Central : weight_mu1_MediumID_Trk_Central) * ((mu2_pt < 30) ? weight_mu2_TightPFIso_TightID_Central : weight_mu2_LoosePFIso_MediumID_Central) * ((mu2_pt < 30) ? weight_mu2_TightID_Trk_Central : weight_mu2_MediumID_Trk_Central) * weight_TrqSF_singleMu_IsoMu24{scale}	weight__singleMuTrigger_{scale}	2024, 2025
weight	Central		weight_gen * weight_lumi * weight_xs * weight_pu * weight_mu1_LoosePFIso_MediumID_Central * weight_mu1_MediumID_Trk_Central * weight_mu2_LoosePFIso_MediumID_Central * weight_mu2_MediumID_Trk_Central * weight_TrqSF_singleMu_IsoMu24Central		2026
weight	PU_up		weight_gen * weight_lumi * weight_xs * weight_pu_up * weight_mu1_LoosePFIso_MediumID_Central * weight_mu1_MediumID_Trk_Central * weight_mu2_LoosePFIso_MediumID_Central * weight_mu2_MediumID_Trk_Central * weight_TrqSF_singleMu_IsoMu24Central	weight__PU_up	2026
weight	PU_down		weight_gen * weight_lumi * weight_xs * weight_pu_down * weight_mu1_LoosePFIso_MediumID_Central * weight_mu1_MediumID_Trk_Central * weight_mu2_LoosePFIso_MediumID_Central * weight_mu2_MediumID_Trk_Central * weight_TrqSF_singleMu_IsoMu24Central	weight__PU_down	2026
weight	MuonID_up		weight_gen * weight_lumi * weight_xs * weight_pu * weight_mu1_LoosePFIso_MediumID_Central * weight_mu1_MediumID_Trk_up * weight_mu2_LoosePFIso_MediumID_Central * weight_mu2_MediumID_Trk_up * weight_TrqSF_singleMu_IsoMu24Central	weight__MuonID_up	2026
weight	MuonID_down		weight_gen * weight_lumi * weight_xs * weight_pu * weight_mu1_LoosePFIso_MediumID_Central * weight_mu1_MediumID_Trk_down * weight_mu2_LoosePFIso_MediumID_Central * weight_mu2_MediumID_Trk_down * weight_TrqSF_singleMu_IsoMu24Central	weight__MuonID_down	2026
weight	MuonIso_up		weight_gen * weight_lumi * weight_xs * weight_pu * weight_mu1_LoosePFIso_MediumID_up * weight_mu1_MediumID_Trk_Central * weight_mu2_LoosePFIso_MediumID_up * weight_mu2_MediumID_Trk_Central * weight_TrqSF_singleMu_IsoMu24Central	weight__MuonIso_up	2026

Continued on next page

Table 5 – continued

Type	Systematic	Muon suffix	Jet suffix / weight expression	Weight column	Eras
weight	MuonIso_down		weight_gen * weight_lumi * weight_xs * weight_pu * weight_mu1_LoosePFIso_MediumID_down * weight_mu1_MediumID_Trk_Central * weight_mu2_LoosePFIso_MediumID_down * weight_mu2_MediumID_Trk_Central * weight_TrpSF_singleMu_IsoMu24Central	weight__MuonIso_down	2026
weight	singleMuTrigger_up		weight_gen * weight_lumi * weight_xs * weight_pu * weight_mu1_LoosePFIso_MediumID_Central * weight_mu1_MediumID_Trk_Central * weight_mu2_LoosePFIso_MediumID_Central * weight_mu2_MediumID_Trk_Central * weight_TrpSF_singleMu_IsoMu24up	weight__singleMuTrigger_up	2026
weight	singleMuTrigger_down		weight_gen * weight_lumi * weight_xs * weight_pu * weight_mu1_LoosePFIso_MediumID_Central * weight_mu1_MediumID_Trk_Central * weight_mu2_LoosePFIso_MediumID_Central * weight_mu2_MediumID_Trk_Central * weight_TrpSF_singleMu_IsoMu24down	weight__singleMuTrigger_down	2026

Table 6: Central event weights used to fill MC histograms.

Era	Central MC weight expression
Run3_2022	genWeight * weight_lumi * weight_xs * weight_pu * weight_mu1_LoosePFIso_MediumID_Central * weight_mu1_MediumID_Trk_Central * weight_mu2_LoosePFIso_MediumID_Central * weight_mu2_MediumID_Trk_Central * weight_TrpSF_singleMu_IsoMu24Central
Run3_2022EE	genWeight * weight_lumi * weight_xs * weight_pu * weight_mu1_LoosePFIso_MediumID_Central * weight_mu1_MediumID_Trk_Central * weight_mu2_LoosePFIso_MediumID_Central * weight_mu2_MediumID_Trk_Central * weight_TrpSF_singleMu_IsoMu24Central
Run3_2023	genWeight * weight_lumi * weight_xs * weight_pu * weight_mu1_LoosePFIso_MediumID_Central * weight_mu1_MediumID_Trk_Central * weight_mu2_LoosePFIso_MediumID_Central * weight_mu2_MediumID_Trk_Central * weight_TrpSF_singleMu_IsoMu24Central
Run3_2023BPix	genWeight * weight_lumi * weight_xs * weight_pu * weight_mu1_LoosePFIso_MediumID_Central * weight_mu1_MediumID_Trk_Central * weight_mu2_LoosePFIso_MediumID_Central * weight_mu2_MediumID_Trk_Central * weight_TrpSF_singleMu_IsoMu24Central
Run3_2024	genWeight * weight_lumi * weight_xs * weight_pu * ((mu1_pt < 30) ? weight_mu1_TightPFIso_TightID_Central : weight_mu1_LoosePFIso_MediumID_Central) * ((mu1_pt < 30) ? weight_mu1_TightID_Trk_Central : weight_mu1_MediumID_Trk_Central) * ((mu2_pt < 30) ? weight_mu2_TightPFIso_TightID_Central : weight_mu2_LoosePFIso_MediumID_Central) * ((mu2_pt < 30) ? weight_mu2_TightID_Trk_Central : weight_mu2_MediumID_Trk_Central) * weight_TrpSF_singleMu_IsoMu24Central
Run3_2025	genWeight * weight_lumi * weight_xs * weight_pu * ((mu1_pt < 30) ? weight_mu1_TightPFIso_TightID_Central : weight_mu1_LoosePFIso_MediumID_Central) * ((mu1_pt < 30) ? weight_mu1_TightID_Trk_Central : weight_mu1_MediumID_Trk_Central) * ((mu2_pt < 30) ? weight_mu2_TightPFIso_TightID_Central : weight_mu2_LoosePFIso_MediumID_Central) * ((mu2_pt < 30) ? weight_mu2_TightID_Trk_Central : weight_mu2_MediumID_Trk_Central) * weight_TrpSF_singleMu_IsoMu24Central
Run3_2026	weight_gen * weight_lumi * weight_xs * weight_pu * weight_mu1_LoosePFIso_MediumID_Central * weight_mu1_MediumID_Trk_Central * weight_mu2_LoosePFIso_MediumID_Central * weight_mu2_MediumID_Trk_Central * weight_TrpSF_singleMu_IsoMu24Central